

Manual Change History (3666184) QST30 G-Drive Electronic Control System Troubleshooting and Repair Manual

Date	Not Available
Section	front
Procedure Number	3666184-toc (/qs3/pubsys2/xml/en/manual/3666184/3666184-toc.html)
Procedure Title	table of contents
Update Reason	

Date	12-Mar-2002
Section	Section i - Introduction
Procedure Number	99-204-004 (/qs3/pubsys2/xml/en/procedures/99/99-204-004.html)
Procedure Title	Symbols
Update Reason	

Date	12-Mar-2002
Section	Section i - Introduction
Procedure Number	99-204-005 (/qs3/pubsys2/xml/en/procedures/99/99-204-005.html)
Procedure Title	Illustrations
Update Reason	

Date	12-Mar-2002
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)
Procedure Title	Component Connector and Pin Inspection
Update Reason	

Date	03-Apr-2002
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-002 (/qs3/pubsys2/xml/en/procedures/99/99-t00-002.html)
Procedure Title	Troubleshooting Symptom Charts
Update Reason	
Date	19-Mar-2003
Section	front
Procedure Number	3666184-titlepage (/qs3/pubsys2/xml/en/manual/3666184/3666184-titlepage.html)
Procedure Title	titlepage
Update Reason	
Date	19-Mar-2003
Section	Section F - Familiarization
Procedure Number	94-101-007 (/qs3/pubsys2/xml/en/procedures/94/94-101-007.html)
Procedure Title	Electronic Controlled Fuel System
Update Reason	
Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc115 (/qs3/pubsys2/xml/en/procedures/94/94-fc115.html)
Procedure Title	Engine Speed Sensor (ESS) Circuit
Update Reason	
Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc135 (/qs3/pubsys2/xml/en/procedures/94/94-fc135.html)
Procedure Title	Oil Pressure Sensor (OPS) Circuit
Update Reason	

Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc141 (/qs3/pubsys2/xml/en/procedures/94/94-fc141.html)
Procedure Title	Oil Pressure Sensor (OPS) Circuit
Update Reason	
Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc143 (/qs3/pubsys2/xml/en/procedures/94/94-fc143.html)
Procedure Title	Oil Pressure - Pre-Low Oil Pressure (LOP) Warning
Update Reason	
Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc144 (/qs3/pubsys2/xml/en/procedures/94/94-fc144.html)
Procedure Title	Coolant Temperature Sensor (CTS) Circuit
Update Reason	
Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc145 (/qs3/pubsys2/xml/en/procedures/94/94-fc145.html)
Procedure Title	Coolant Temperature Sensor (CTS) Circuit
Update Reason	
Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc146 (/qs3/pubsys2/xml/en/procedures/94/94-fc146.html)
Procedure Title	Coolant Temperature - Engine Protection
Update Reason	

Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc151 (/qs3/pubsys2/xml/en/procedures/94/94-fc151.html)
Procedure Title	Coolant Temperature - Engine Protection
Update Reason	
Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc171 (/qs3/pubsys2/xml/en/procedures/94/94-fc171.html)
Procedure Title	Rack Position
Update Reason	
Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc234 (/qs3/pubsys2/xml/en/procedures/94/94-fc234.html)
Procedure Title	Engine Overspeed - Engine Protection
Update Reason	
Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc342 (/qs3/pubsys2/xml/en/procedures/94/94-fc342.html)
Procedure Title	Electronic Control Module (ECM) Memory - Area Faults
Update Reason	
Date	19-Mar-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	94-fc415 (/qs3/pubsys2/xml/en/procedures/94/94-fc415.html)
Procedure Title	Oil Pressure - Engine Protection
Update Reason	

Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-022-015 (/qs3/pubsys2/xml/en/procedures/94/94-022-015.html)
Procedure Title	Wiring Harness Repair Kit
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-006 (/qs3/pubsys2/xml/en/procedures/94/94-019-006.html)
Procedure Title	Data Link, Service Tool
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-014 (/qs3/pubsys2/xml/en/procedures/94/94-019-014.html)
Procedure Title	Stop/Run Switch
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-015 (/qs3/pubsys2/xml/en/procedures/94/94-019-015.html)
Procedure Title	Stop/Run Switch Circuit
Update Reason	

Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-019 (/qs3/pubsys2/xml/en/procedures/94/94-019-019.html)
Procedure Title	Engine Coolant Temperature Sensor
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-031 (/qs3/pubsys2/xml/en/procedures/94/94-019-031.html)
Procedure Title	Electronic Control Module (ECM)
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-032 (/qs3/pubsys2/xml/en/procedures/94/94-019-032.html)
Procedure Title	ECM Calibration Code
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-042 (/qs3/pubsys2/xml/en/procedures/94/94-019-042.html)
Procedure Title	Engine Speed Sensor (ESS)
Update Reason	

Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-043 (/qs3/pubsys2/xml/en/procedures/94/94-019-043.html)
Procedure Title	Engine Wiring Harness
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-049 (/qs3/pubsys2/xml/en/procedures/94/94-019-049.html)
Procedure Title	Fuel Shutoff Valve Circuit
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-050 (/qs3/pubsys2/xml/en/procedures/94/94-019-050.html)
Procedure Title	Fuel Shutoff Valve
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-066 (/qs3/pubsys2/xml/en/procedures/94/94-019-066.html)
Procedure Title	Engine Oil Pressure Sensor
Update Reason	

Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-087 (/qs3/pubsys2/xml/en/procedures/94/94-019-087.html)
Procedure Title	Unswitched Battery Supply Circuit
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-095 (/qs3/pubsys2/xml/en/procedures/94/94-019-095.html)
Procedure Title	Idle Rated Switch
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-096 (/qs3/pubsys2/xml/en/procedures/94/94-019-096.html)
Procedure Title	Idle Rated Switch Circuit
Update Reason	
Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-106 (/qs3/pubsys2/xml/en/procedures/94/94-019-106.html)
Procedure Title	Engine Speed Sensor Circuit
Update Reason	

Date	24-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-019-240 (/qs3/pubsys2/xml/en/procedures/94/94-019-240.html)
Procedure Title	Connector, 40-Pin
Update Reason	
Date	24-Mar-2003
Section	Section L - Service Literature
Procedure Number	94-205-001 (/qs3/pubsys2/xml/en/procedures/94/94-205-001.html)
Procedure Title	Additional Service Literature
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-197 (/qs3/pubsys2/xml/en/procedures/99/99-019-197.html)
Procedure Title	Ring Terminal
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-199 (/qs3/pubsys2/xml/en/procedures/99/99-019-199.html)
Procedure Title	Connector, Butt Splice
Update Reason	

Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html)
Procedure Title	Metripack Connector Series
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-203 (/qs3/pubsys2/xml/en/procedures/99/99-019-203.html)
Procedure Title	AMP Connector Series
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html)
Procedure Title	Deutsch DRC Connector Series
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-209 (/qs3/pubsys2/xml/en/procedures/99/99-019-209.html)
Procedure Title	ITT Cannon Connector Series
Update Reason	

Date	09-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	94-022-001_19 (/qs3/pubsys2/xml/en/procedures/94/94-022-001_19.html)
Procedure Title	Service Tools
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-010 (/qs3/pubsys2/xml/en/procedures/99/99-018-010.html)
Procedure Title	Fraction, Decimal, Millimeter Conversions
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-011 (/qs3/pubsys2/xml/en/procedures/99/99-018-011.html)
Procedure Title	Newton-Meter to Foot-Pound Conversions
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-012 (/qs3/pubsys2/xml/en/procedures/99/99-018-012.html)
Procedure Title	Pipe Plug Torque Values
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-013 (/qs3/pubsys2/xml/en/procedures/99/99-018-013.html)
Procedure Title	Tap-Drill Chart - U.S. Customary and Metric
Update Reason	

Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-014 (/qs3/pubsys2/xml/en/procedures/99/99-018-014.html)
Procedure Title	Weights and Measures - Conversion Factors
Update Reason	
Date	22-May-2006
Section	front
Procedure Number	tr-frwd (/qs3/pubsys2/xml/en/manual/preface/tr-frwd.html)
Procedure Title	foreword
Update Reason	
Date	27-Jun-2006
Section	Section L - Service Literature
Procedure Number	99-205-002 (/qs3/pubsys2/xml/en/procedures/99/99-205-002.html)
Procedure Title	Service Literature Ordering Location
Update Reason	
Date	24-Jul-2007
Section	Section i - Introduction
Procedure Number	99-204-008 (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)
Procedure Title	General Cleaning Instructions
Update Reason	
Date	07-Nov-2007
Section	Section V - Specifications
Procedure Number	3666185 (/qs3/pubsys2/xml/en/procedures//3666185.html)
Procedure Title	QST30 G-Drive Wiring Diagram Specifications
Update Reason	

Date	28-Jan-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)
Procedure Title	Resistance Measurement Using a Multimeter and a Wiring Diagram
Update Reason	

Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t016-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t016-tr.html)
Procedure Title	Communications Error – Electronic Service Tool or Control Device
Update Reason	

Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t033-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t033-tr.html)
Procedure Title	Engine Acceleration or Response Slow
Update Reason	

Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t041-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t041-tr.html)
Procedure Title	Engine Decelerates Slowly
Update Reason	

Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t043-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t043-tr.html)
Procedure Title	Engine Difficult to Start or Will Not Start (Exhaust Smoke)
Update Reason	
Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t044-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t044-tr.html)
Procedure Title	Engine Difficult to Start or Will Not Start (No Exhaust Smoke)
Update Reason	
Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t057-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t057-tr.html)
Procedure Title	Engine Power Output Low
Update Reason	
Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t062-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t062-tr.html)
Procedure Title	Engine Runs Rough or Misfires
Update Reason	

Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t064-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t064-tr.html)
Procedure Title	Engine Shuts Off Unexpectedly or Dies During Deceleration
Update Reason	
Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t066-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t066-tr.html)
Procedure Title	Engine Speed Surges at Low or High Idle
Update Reason	
Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t080-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t080-tr.html)
Procedure Title	Engine Will Not Reach Rated Speed (RPM)
Update Reason	
Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t087-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t087-tr.html)
Procedure Title	Fuel Consumption Excessive
Update Reason	
Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t091-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t091-tr.html)
Procedure Title	Fuel in Coolant
Update Reason	

Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t092-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t092-tr.html)
Procedure Title	Fuel in Lubricating Oil
Update Reason	
Date	06-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t116-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t116-tr.html)
Procedure Title	Smoke, Black-Excessive
Update Reason	
Date	06-Feb-2008
Section	back
Procedure Number	3666184-backcover (/qs3/pubsys2/xml/en/manual/3666184/3666184-backcover.html)
Procedure Title	Backcover
Update Reason	
Date	14-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t118-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t118-tr.html)
Procedure Title	Smoke, White-Excessive
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-008 (/qs3/pubsys2/xml/en/procedures/99/99-019-008.html)
Procedure Title	Battery Ground Circuit
Update Reason	

Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-047 (/qs3/pubsys2/xml/en/procedures/99/99-019-047.html)
Procedure Title	Fault Lamp Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-071 (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)
Procedure Title	OEM Wiring Harness
Update Reason	
Date	16-Dec-2008
Section	Section E - Engine Identification
Procedure Number	94-100-001 (/qs3/pubsys2/xml/en/procedures/94/94-100-001.html)
Procedure Title	Engine Identification
Update Reason	
Date	16-Dec-2008
Section	Section E - Engine Identification
Procedure Number	94-100-002 (/qs3/pubsys2/xml/en/procedures/94/94-100-002.html)
Procedure Title	Engine Diagrams
Update Reason	
Date	25-Jun-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-001-tr (/qs3/pubsys2/xml/en/procedures/99/99-t00-001-tr.html)
Procedure Title	Troubleshooting Procedures and Techniques
Update Reason	

Date	22-Jul-2009
Section	Section L - Service Literature
Procedure Number	99-205-004 (/qs3/pubsys2/xml/en/procedures/99/99-205-004.html)
Procedure Title	Cummins Customized Parts Catalog
Update Reason	
Date	10-Sep-2009
Section	Section i - Introduction
Procedure Number	99-204-002-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-002-tr.html)
Procedure Title	About the Manual
Update Reason	
Date	08-Oct-2009
Section	Section i - Introduction
Procedure Number	99-204-007-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-007-tr.html)
Procedure Title	General Repair Instructions
Update Reason	
Date	02-Dec-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)
Procedure Title	Multimeter Usage
Update Reason	
Date	02-Dec-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)
Procedure Title	Inactive or Intermittent Fault Code
Update Reason	

Date	25-Jan-2010
Section	Section i - Introduction
Procedure Number	99-204-006-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-006-tr.html)
Procedure Title	General Safety Instructions
Update Reason	
Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-005 (/qs3/pubsys2/xml/en/procedures/99/99-018-005.html)
Procedure Title	Drive Belt Tension
Update Reason	
Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-009 (/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)
Procedure Title	Capscrew Markings and Torque Values
Update Reason	
Date	24-Mar-2010
Section	Section i - Introduction
Procedure Number	99-204-009 (/qs3/pubsys2/xml/en/procedures/99/99-204-009.html)
Procedure Title	Acronyms and Abbreviations
Update Reason	
Date	25-May-2010
Section	Section i - Introduction
Procedure Number	99-204-003-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-003-tr.html)
Procedure Title	How to Use the Manual
Update Reason	

Date	17-Apr-2014
Section	Section 19 - Electronic Controls
Procedure Number	94-019-031 (/qs3/pubsys2/xml/en/procedures/94/94-019-031.html)
Procedure Title	Engine Control Module
Update Reason	Revise jackscrew torque value and added a note to not over torque.

Date	17-Apr-2014
Section	Section 19 - Electronic Controls
Procedure Number	94-019-031 (/qs3/pubsys2/xml/en/procedures/94/94-019-031.html)
Procedure Title	Engine Wiring Harness
Update Reason	Revise jackscrew torque value and added a note to not over torque.

Date	11-Jul-2025
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666184-t092-5-tr (/qs3/pubsys2/xml/en/manual/3666184/3666184-t092-5-tr.html)
Procedure Title	Fuel Leak - External
Update Reason	Added troubleshooting symptom tree to manual.

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